

NWB NEURODATA WITHOUT BORDERS

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Data heterogeneity in neurophysiology

- **Proprietary acquisition formats**

Every recording system (electrophysiology, optical imaging, behavior tracking) writes its own binary layout.

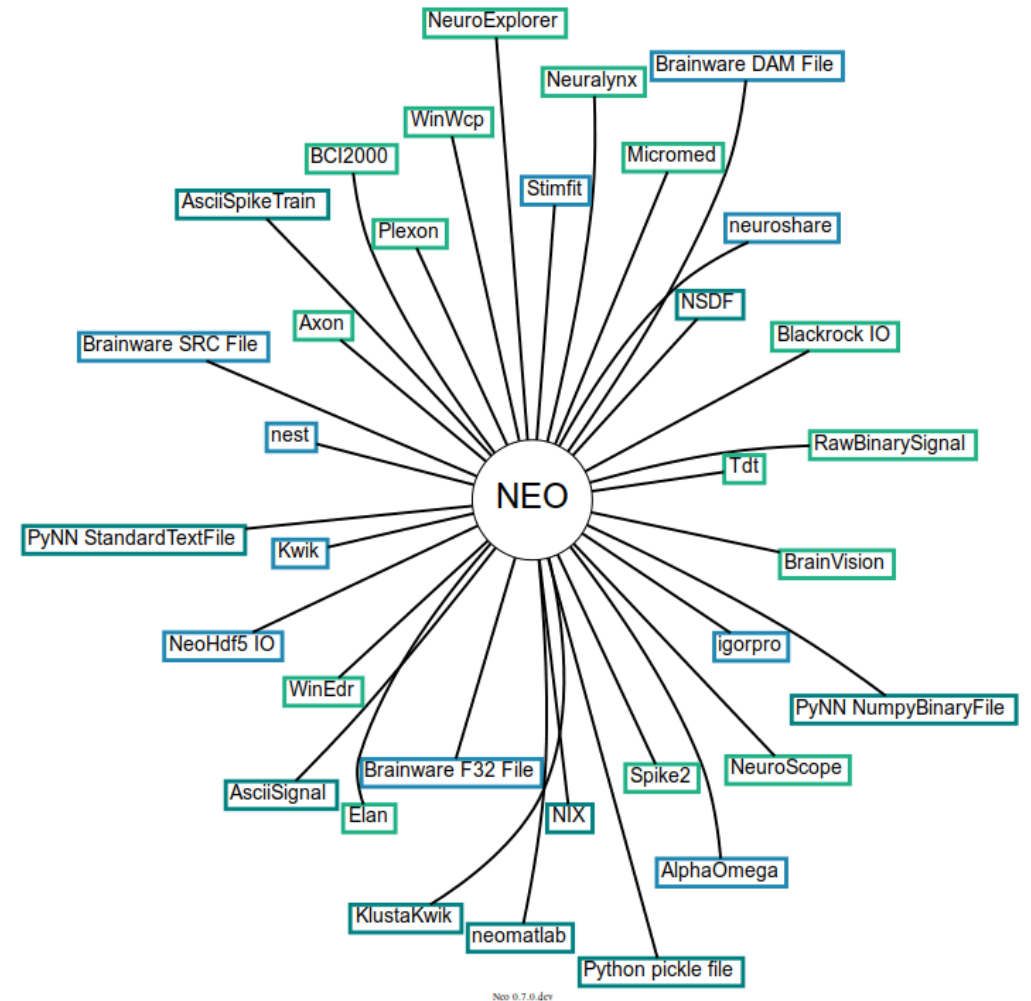
Proprietary acquisition formats: Problems

The problem:

- Hardware vendors invent their own file formats → **large diversity**
- Proprietary, with little to no official documentation → lacks **rigor**
- Neo maintains the largest collection of community-driven I/Os, → **huge maintenance costs** (e.g., new versions)

Why?

- A **file format (=sequence of data on disk)** is not necessarily based on a good **data representation**:
 - Vendors need to write out large data fast → file format is optimized for this task
 - Vendors concentrate on metadata related to their setup
- \$\$\$



Proprietary acquisition formats: Solutions

The past:

Neuroshare (failed)

- Launched in the early 2000s to create a common API and data format
- MATLAB based approach to bring together data acquired from different hardware manufacturers (e.g., Plexon, Tucker-Davis Technologies, Alpha Omega) by a common API
- Windows DLLs (Dynamic Link Libraries) for MATLAB provided by vendors

The present:

OpenEphys: Completely open and modular hardware platform with open formats

Neo: Common, flexible data representation with corresponding readers, community-developed

NWB: Common schemas and APIs for storing electrophysiological data along with tooling support

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- **Inconsistent metadata conventions**

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Inconsistent metadata conventions

Only recently, many initiatives have evolved to develop comprehensive and interoperable metadata vocabularies, schemas, ontologies:

G-Node terminologies: Specific early attempt to cover electrophysiology-related terms

<https://terminologies.g-node.org/v1.1/terminologies.xml>

Interlex: Crowd-sourced definitions for neuroscience-related terms

<https://scicrunch.org/scicrunch/interlex/dashboard>

openMINDS: Initiative of EBRAINS to develop schemas covering descriptions of various parts of experimental and computational neuroscience data

<https://wiki.ebrains.eu/bin/view/Collabs/openminds/>

NWB: Specific approach to precisely define the aspects related to electrophysiological recordings and their processing.

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Re-analysis or cross-lab comparison requires custom parsers for each source format.

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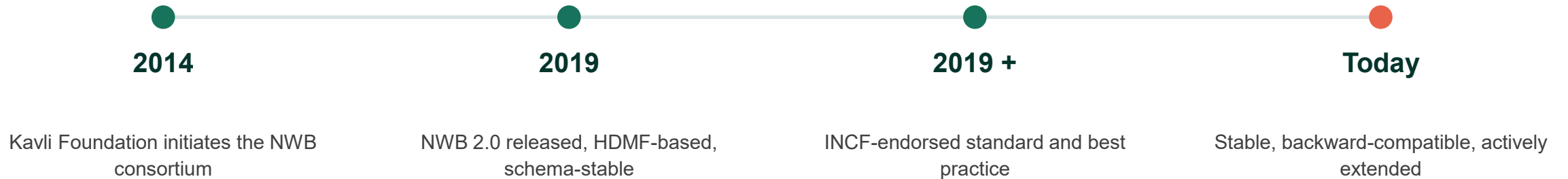
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- **Limited long-term reusability**

Without a shared schema, archived data becomes difficult to interpret once tribal knowledge is lost.

Evolution of NWB

NWB is a community data standard for neurophysiology: a schema, a set of APIs, and an ecosystem of tools, rather than a single fixed file format.



GOVERNING BODY

Schema maintained via the open NWB Specification Language

SCOPE

Intracellular & extracellular ephys, optical physiology, behavior

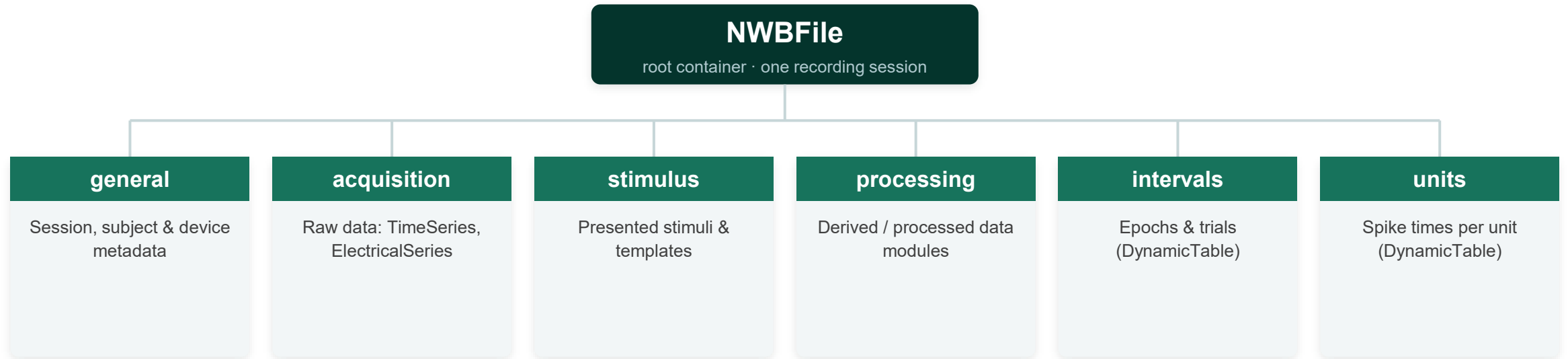
STATUS

Schema-stable since v2.0, strictly backward-compatible

Design principles

- **Schema-driven:** The NWB Specification Language (YAML) formally defines every neurodata_type: groups, datasets, attributes, and links.
- **Built on HDMF:** The Hierarchical Data Modeling Framework provides the common schema language and I/Os shared across science domains.
- **Hierarchical container model:** Data and metadata live together in one nested structure, analogous to a typed, self-describing filesystem.
- **Storage-format agnostic:** The same logical schema maps onto either an HDF5 file or a Zarr store, decoupling data model from storage backend.
- **Extensible by design:** Labs define custom neurodata_types as namespace extensions (ndx-*) without forking the core schema.

NWBFile: hierarchical structure



All six branches share the same container model: groups, typed datasets, and attributes, addressable by path within the file.

Core neurodata_types



TimeSeries

Base type for any time-stamped data array; nearly every measurement in NWB derives from it.



ElectricalSeries

Extracellular voltage traces recorded from one or more electrodes.



SpikeEventSeries

Detected spike waveforms and their timestamps.



Units

Spike times and per-neuron metadata, stored as a DynamicTable.



DynamicTable

Generic tabular container for metadata: electrodes, units, trials.



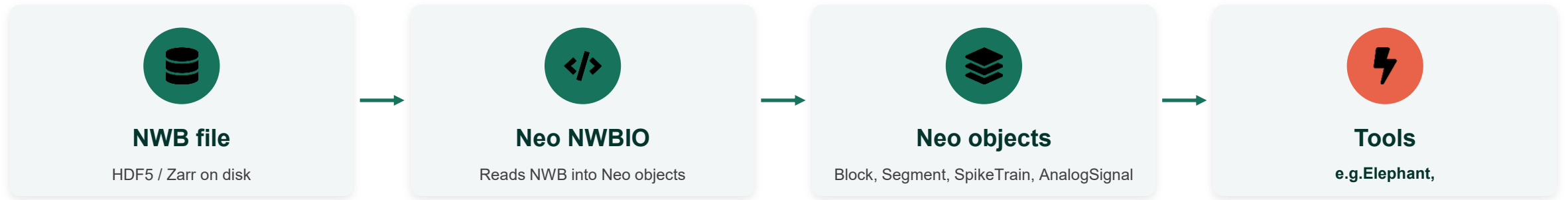
TimeIntervals

Labeled time windows such as trials or behavioral epochs.

APIs and tooling

- **PyNWB:** Reference Python API for reading, writing, and validating NWB files.
- **MatNWB:** MATLAB API offering equivalent read/write functionality.
- **HDMF:** Underlying schema language and I/O layer shared across NWB and other HDMF-based standards.
- **NWB Inspector:** Static checker for schema compliance and best-practice metadata conventions.
- **Extension catalog (ndx-):** Community-maintained repository of namespace extensions for specialized data types.

From NWB file to analysis



- Avoids bespoke parsers per acquisition system in analysis pipelines
- Keeps electrode, unit, and trial metadata attached to derived results
- Supports reproducible, re-runnable workflows on archived datasets

Summary

- **Schema-driven, hierarchical standard** for neurophysiology data
- **Storage backends**: HDF5 for local files, Zarr for cloud-scale workflows
- Backed by a stable **API ecosystem**: PyNWB, MatNWB, HDMF, NWB Inspector
- Supports **FAIR sharing** via the DANDI Archive or the EBRAINS Knowledgegraph and re-use by Neo-based analysis tools

Resources

Project home: nwb.org

pyNWB documentation: pynwb.readthedocs.io